

U24AI050 LAB 1

08 January 2026 13:07

1) Study network configuration commands and execute them on your system. For Example:

1. IPCONFIG

ipconfig

ipconfig /all

```
Command Prompt
Microsoft Windows [Version 10.0.26200.7462]
(c) Microsoft Corporation. All rights reserved.

C:\Users\NEHA>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::dacf:818a:80f8:32af%51
    IPv4 Address. . . . . : 192.168.56.1
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 

C:\Users\NEHA>ipconfig /all

Windows IP Configuration

    Host Name . . . . . : krishna
    Primary Dns Suffix . . . . . : 
    Node Type . . . . . : Mixed
    IP Routing Enabled. . . . . : No
    WINS Proxy Enabled. . . . . : No

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Description . . . . . : VirtualBox Host-Only Ethernet Adapter
    Physical Address. . . . . : 0A-0B-27-00-00-33
    DHCP Enabled. . . . . : No
    Autoconfiguration Enabled . . . . : Yes
    Link-local IPv6 Address . . . . . : fe80::dacf:818a:80f8:32af%51(Preferred)
    IPv4 Address. . . . . : 192.168.56.1(Preferred)
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : 
    DHCPv6 IAID . . . . . : 856293015
    DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-C5-D0-00-00-00-16-01-02-0E
    NetBIOS over Lcpip. . . . . : Enabled

Unknown adapter McAfee VPN:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
    Description . . . . . : TAP-Windows Adapter V9
    Physical Address. . . . . : 00-FF-0C-67-A0-92
    DHCP Enabled. . . . . : Yes
    Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix  . : 
    Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
```

```
Command Prompt
Wireless LAN adapter Wi-Fi:
Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Wi-Fi 6 AX203
Physical Address. . . . . : 49-A3-BB-FE-C8-A8
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::8a5e:2c19:8467:6752%28(Preferred)
IPv4 Address. . . . . : 172.33.8.16(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : 68 January 2026 18:28:16
Lease Expires . . . . . : 68 January 2026 16:19:17
Default Gateway . . . . . : 172.33.8.1
DHCP Server . . . . . : 172.16.1.2
DHCPv6 IAD . . . . . : 306489251
DHCPv6 Client DUID. . . . . : 00-01-00-01-2E-C5-00-00-00-10-01-42-6E
DNS Servers . . . . . : 8.8.8.8
                        172.16.1.1
NetBIOS over Tcpip. . . . . : Enabled

Ethernet adapter Bluetooth Network Connection:
Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Bluetooth Device (Personal Area Network)
Physical Address. . . . . : 40-A3-BB-FE-C8-AC
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

C:\Users\NEHA>nslookup google.com
Server: dns.google
Address: 8.8.8.8

Non-authoritative answer:
Name: google.com
Addresses: 2404:6800:4000:817::200e
          142.250.190.14

C:\Users\NEHA>
```

2. NSLOOKUP
 nslookup google.com
3. HOSTNAME
 hostname
4. PING
 ping google.com
 ping 8.8.8.8

```
Command Prompt
C:\Users\NEHA>nslookup google.com
Server:  dns.google
Address:  8.8.8.8

Non-authoritative answer:
Name:     google.com
Addresses: 2404:6800:4009:817::200e
          142.250.195.14

C:\Users\NEHA>hostname
krishna

C:\Users\NEHA>ping google.com

Pinging google.com [142.250.195.14] with 32 bytes of data:
Reply from 142.250.195.14: bytes=32 time=26ms TTL=116
Reply from 142.250.195.14: bytes=32 time=12ms TTL=116
Reply from 142.250.195.14: bytes=32 time=10ms TTL=116
Reply from 142.250.195.14: bytes=32 time=19ms TTL=116

Ping statistics for 142.250.195.14:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 10ms, Maximum = 26ms, Average = 16ms

C:\Users\NEHA>ping 8.8.8.8

Pinging 8.8.8.8 with 32 bytes of data:
Reply from 8.8.8.8: bytes=32 time=24ms TTL=115
Reply from 8.8.8.8: bytes=32 time=25ms TTL=115
Reply from 8.8.8.8: bytes=32 time=36ms TTL=115
Reply from 8.8.8.8: bytes=32 time=45ms TTL=115

Ping statistics for 8.8.8.8:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 24ms, Maximum = 45ms, Average = 32ms

C:\Users\NEHA>
```

5. TRACERT

tracert google.com

6. NETSTAT

netstat

netstat -an

netstat -r

7. ARP(Address Resolution Protocol)

arp -a

```
C:\Users\NEHA>tracert google.com
```

```
Tracing route to google.com [142.250.70.46]  
over a maximum of 30 hops:
```

```
  1      *          *          *      Request timed out.  
  2      3 ms      2 ms      4 ms    192.168.1.1  
  3      *          3 ms      2 ms    172.16.1.1  
  4      4 ms      3 ms      2 ms    192.168.168.1  
  5      8 ms      8 ms      14 ms   160.202.10.65  
  6      *          4 ms      4 ms    10.194.29.73  
  7      8 ms      7 ms      7 ms    10.250.10.121  
  8      ^C
```

```
C:\Users\NEHA>netstat
```

```
Active Connections
```

Proto	Local Address	Foreign Address	State
TCP	127.0.0.1:49714	krishna:49715	ESTABLISHED
TCP	127.0.0.1:49715	krishna:49714	ESTABLISHED
TCP	127.0.0.1:49716	krishna:49717	ESTABLISHED
TCP	127.0.0.1:49717	krishna:49716	ESTABLISHED
TCP	127.0.0.1:49719	krishna:49720	ESTABLISHED
TCP	127.0.0.1:49720	krishna:49719	ESTABLISHED
TCP	127.0.0.1:54802	krishna:54803	ESTABLISHED
TCP	127.0.0.1:54803	krishna:54802	ESTABLISHED
TCP	172.33.8.16:49666	104.18.39.21:https	ESTABLISHED
TCP	172.33.8.16:49667	172.66.154.160:https	ESTABLISHED
TCP	172.33.8.16:49821	4.213.25.240:https	ESTABLISHED

```
C:\Users\NEHA>netstat -an
```

Active Connections

Proto	Local Address	Foreign Address	State
TCP	0.0.0.0:135	0.0.0.0:0	LISTENING
TCP	0.0.0.0:445	0.0.0.0:0	LISTENING
TCP	0.0.0.0:5040	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49668	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49669	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49672	0.0.0.0:0	LISTENING
TCP	0.0.0.0:49687	0.0.0.0:0	LISTENING
TCP	0.0.0.0:50131	0.0.0.0:0	LISTENING
TCP	127.0.0.1:49714	127.0.0.1:49715	ESTABLISHED
TCP	127.0.0.1:49715	127.0.0.1:49714	ESTABLISHED
TCP	127.0.0.1:49716	127.0.0.1:49717	ESTABLISHED
TCP	127.0.0.1:49717	127.0.0.1:49716	ESTABLISHED
TCP	127.0.0.1:49719	127.0.0.1:49720	ESTABLISHED
TCP	127.0.0.1:49720	127.0.0.1:49719	ESTABLISHED
TCP	127.0.0.1:54802	127.0.0.1:54803	ESTABLISHED
TCP	127.0.0.1:54803	127.0.0.1:54802	ESTABLISHED
TCP	172.33.8.16:139	0.0.0.0:0	LISTENING
TCP	172.33.8.16:49666	104.18.39.21:443	ESTABLISHED
TCP	172.33.8.16:49667	172.66.154.160:443	ESTABLISHED
TCP	172.33.8.16:49670	8.8.4.4:443	SYN_SENT
TCP	172.33.8.16:49821	4.213.25.240:443	ESTABLISHED
TCP	172.33.8.16:49822	4.213.25.240:443	ESTABLISHED
TCP	172.33.8.16:49980	204.79.197.222:443	ESTABLISHED
TCP	172.33.8.16:50060	104.18.39.21:443	ESTABLISHED
TCP	172.33.8.16:51044	104.18.8.59:443	ESTABLISHED
TCP	172.33.8.16:51525	13.107.19.254:443	ESTABLISHED
TCP	172.33.8.16:52055	104.18.32.47:443	ESTABLISHED
TCP	172.33.8.16:52603	104.21.49.153:443	ESTABLISHED
TCP	172.33.8.16:52860	172.217.194.188:5228	ESTABLISHED
TCP	172.33.8.16:54138	104.26.7.247:443	ESTABLISHED
TCP	172.33.8.16:54247	104.18.39.21:443	ESTABLISHED

```

UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5353      *: *
UDP    0.0.0.0:5355      *: *
UDP    0.0.0.0:54274     142.251.221.227:443
UDP    0.0.0.0:59948      *: *
UDP    0.0.0.0:60899      150.171.22.12:443
UDP    0.0.0.0:62514      23.218.249.153:443
UDP    127.0.0.1:1900      *: *
UDP    127.0.0.1:60908      *: *
UDP    127.0.0.1:62967     127.0.0.1:62967
UDP    172.33.8.16:137      *: *
UDP    172.33.8.16:138      *: *
UDP    172.33.8.16:1900      *: *
UDP    172.33.8.16:60907      *: *
UDP    192.168.56.1:137      *: *
UDP    192.168.56.1:138      *: *
UDP    192.168.56.1:1900      *: *
UDP    192.168.56.1:60906      *: *
UDP    [::]:5353            *: *
UDP    [::]:5353            *: *
UDP    [::]:5353            *: *
UDP    [::]:5353            *: *
UDP    [::]:5353            *: *
UDP    [::]:5355            *: *
UDP    [::]:59948          *: *
UDP    [::1]:1900           *: *
UDP    [::1]:60905          *: *
UDP    [fe80::8a5e:2c19:8467:6752%20]:1900  *: *
UDP    [fe80::8a5e:2c19:8467:6752%20]:60904  *: *
UDP    [fe80::dacf:818a:80f8:32af%51]:1900  *: *
UDP    [fe80::dacf:818a:80f8:32af%51]:60903  *: *

```

```

C:\Users\NEHA>netstat -r
=====
Interface List
51...0a 00 27 00 00 33 .....VirtualBox Host-Only Ethernet Adapter
4...00 ff 0c 67 a0 92 .....TAP-Windows Adapter V9
14...44 a3 bb fe cb a9 .....Microsoft Wi-Fi Direct Virtual Adapter
9...46 a3 bb fe cb a8 .....Microsoft Wi-Fi Direct Virtual Adapter #2
20...44 a3 bb fe cb a8 .....Intel(R) Wi-Fi 6 AX203
16...44 a3 bb fe cb ac .....Bluetooth Device (Personal Area Network)
1.....Software Loopback Interface 1
=====

IPv4 Route Table
=====
Active Routes:
Network Destination        Netmask          Gateway           Interface        Metric
0.0.0.0                    0.0.0.0          172.33.8.1        172.33.8.16      60
127.0.0.0                  255.0.0.0        On-link           127.0.0.1        331
127.0.0.1                  255.255.255.255  On-link           127.0.0.1        331
127.255.255.255            255.255.255.255  On-link           127.0.0.1        331
172.33.8.0                  255.255.255.0    On-link           172.33.8.16      316
172.33.8.16                 255.255.255.255  On-link           172.33.8.16      316
172.33.8.255                255.255.255.255  On-link           172.33.8.16      316
192.168.56.0                255.255.255.0    On-link           192.168.56.1     330
192.168.56.1                255.255.255.255  On-link           192.168.56.1     330
192.168.56.255              255.255.255.255  On-link           192.168.56.1     330
224.0.0.0                   240.0.0.0        On-link           127.0.0.1        331
224.0.0.0                   240.0.0.0        On-link           172.33.8.16      316
224.0.0.0                   240.0.0.0        On-link           192.168.56.1     330
255.255.255.255             255.255.255.255  On-link           127.0.0.1        331
255.255.255.255             255.255.255.255  On-link           172.33.8.16      316
255.255.255.255             255.255.255.255  On-link           192.168.56.1     330
=====
Persistent Routes:
None

IPv6 Route Table
=====
Active Routes:
If Metric Network Destination      Gateway
1      331 ::1/128                On-link

```

IPv6 Route Table

Active Routes:

If	Metric	Network	Destination	Gateway
1	331	::1/128		On-link
20	316	fe80::/64		On-link
51	281	fe80::/64		On-link
20	316	fe80::8a5e:2c19:8467:6752/128		On-link
51	281	fe80::dacf:818a:80f8:32af/128		On-link
1	331	ff00::/8		On-link
20	316	ff00::/8		On-link
51	281	ff00::/8		On-link

Persistent Routes:

None

C:\Users\NEHA>arp -a

Interface: 172.33.8.16 --- 0x14

Internet Address	Physical Address	Type
172.33.8.1	38-22-d6-9c-fb-59	dynamic
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

Interface: 192.168.56.1 --- 0x33

Internet Address	Physical Address	Type
192.168.56.255	ff-ff-ff-ff-ff-ff	static
224.0.0.22	01-00-5e-00-00-16	static
224.0.0.251	01-00-5e-00-00-fb	static
224.0.0.252	01-00-5e-00-00-fc	static
239.255.255.250	01-00-5e-7f-ff-fa	static
255.255.255.255	ff-ff-ff-ff-ff-ff	static

C:\Users\NEHA>

8. SYSTEMINFO

systeminfo

```
C:\Users\NEHA>systeminfo

Host Name:                KRISHNA
OS Name:                  Microsoft Windows 11 Home Single Language
OS Version:               10.0.26200 N/A Build 26200
OS Manufacturer:         Microsoft Corporation
OS Configuration:        Standalone Workstation
OS Build Type:             Multiprocessor Free
Registered Owner:         NEHA
Registered Organization:   N/A
Product ID:               00342-42724-01102-AAOEM
Original Install Date:    13-03-2025, 09:11:03
System Boot Time:         08-01-2026, 08:53:32
System Manufacturer:      LENOVO
System Model:              83EM
System Type:              x64-based PC
Processor(s):              1 Processor(s) Installed.
                           [01]: Intel64 Family 6 Model 186 Stepping 2 GenuineIntel ~2400 Mhz
BIOS Version:              LENOVO LTCN34WW, 16-07-2024
Windows Directory:        C:\Windows
System Directory:          C:\Windows\system32
Boot Device:               \Device\HarddiskVolume1
System Locale:              en-us;English (United States)
Input Locale:              00004009
Time Zone:                 (UTC+05:30) Chennai, Kolkata, Mumbai, New Delhi
Total Physical Memory:     16,077 MB
Available Physical Memory: 6,975 MB
Virtual Memory: Max Size: 21,965 MB
Virtual Memory: Available: 11,125 MB
Virtual Memory: In Use:    10,840 MB
Page File Location(s):     C:\pagefile.sys
Domain:                    WORKGROUP
Logon Server:              \\KRISHNA
Hotfix(s):                  4 Hotfix(s) Installed.
                           [01]: KB5067931
                           [02]: KB5054156
                           [03]: KB5072033
                           [04]: KB5071142
Network Card(s):           4 NIC(s) Installed.
                           [01]: Intel(R) Wi-Fi 6 AX203
```

```

Network Card(s):      4 NIC(s) Installed.
                      [01]: Intel(R) Wi-Fi 6 AX203
                        Connection Name: Wi-Fi
                        DHCP Enabled: Yes
                        DHCP Server: 172.16.1.2
                        IP address(es)
                        [01]: 172.33.8.16
                        [02]: fe80::8a5e:2c19:8467:6752
                      [02]: Bluetooth Device (Personal Area Network)
                        Connection Name: Bluetooth Network Connection
                        Status: Media disconnected
                      [03]: TAP-Windows Adapter V9
                        Connection Name: McAfee VPN
                        Status: Media disconnected
                      [04]: VirtualBox Host-Only Ethernet Adapter
                        Connection Name: Ethernet 3
                        DHCP Enabled: No
                        IP address(es)
                        [01]: 192.168.56.1
                        [02]: fe80::dacf:818a:80f8:32af

Virtualization-based security: Status: Running
                                Required Security Properties:
                                    Base Virtualization Support
                                Available Security Properties:
                                    Base Virtualization Support
                                    Secure Boot
                                    DMA Protection
                                    UEFI Code Readonly
                                    SMM Security Mitigations 1.0
                                    Mode Based Execution Control
                                    APIC Virtualization
                                Services Configured:
                                    Hypervisor enforced Code Integrity
                                Services Running:
                                    Hypervisor enforced Code Integrity
                                App Control for Business policy: Enforced
                                App Control for Business user mode policy: Off
                                Security Features Enabled:

Hyper-V Requirements:  A hypervisor has been detected. Features required for Hyper-V will not be displayed.

```

2) Explore computer network setup.

Introduction

A computer network setup is the arrangement of hardware, software, and communication links that allow multiple computers and devices to communicate and share resources such as data, printers, and internet access.

Basic Components of a Network

1. End Devices

- Computer, Laptop, Mobile, Printer
- Used by users to send and receive data

2. Networking Devices

- **Router:** Connects local network to the internet
- **Switch:** Connects multiple devices within a LAN
- **Modem:** Converts ISP signals to digital data

3. Transmission Media

- Wired: Ethernet cable, Fiber optics
- Wireless: Wi-Fi, Bluetooth

4. Network Protocols

- TCP/IP, HTTP, HTTPS, FTP, DNS
- Define rules for data communication

Types of Network Setup

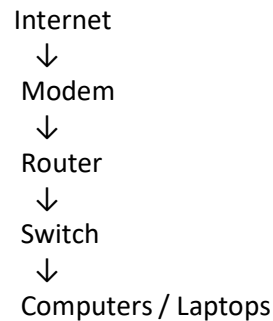
Network Type	Description
--------------	-------------

LAN	Used in homes, labs, colleges
WAN	Covers large geographical area (Internet)
MAN	City-wide network
PAN	Personal devices network

Common Network Topologies

1. **Star Topology** – All devices connected to a central switch
2. **Bus Topology** – All devices connected to a single cable
3. **Ring Topology** – Devices connected in circular manner
4. **Mesh Topology** – Each device connected to every other device

Typical Computer Network Setup (Home / Lab)



Advantages of Computer Network

- Resource sharing
- Fast communication
- Centralized data management
- Internet sharing

Conclusion

In this experiment, we explored computer network setup including network components, types of networks, topologies, and a typical home or laboratory network structure.